# Quantum Computing, Quantum Machine Learning and Quantum Information Theories

Morten Hjorth-Jensen[1,2]

Department of Physics, University of Oslo[1]

Department of Physics and Astronomy and Facility for Rare Isotope Beams, Michigan State University, USA[2]

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# Plans for the week of February 10-14

1. Reminder from last week on gates and circuits
2. One-qubit and two-qubit gates, background and realizations
3. Entropy as a measurement of entanglement
4. Simple Hamiltonian systems and how to use the density matrix to estimate degrees of entanglement
5. Video of lecture to be added

# Quantum gates, circuits and simple algorithms

Quantum gates are physical actions that are applied to the physical system representing the qubits. Mathematically, they are complex-valued, unitary matrices which act on the complex-values normalized vectors that represent qubits. As the quantum analog of classical logic gates (such as AND and OR), there is a corresponding quantum gate for every classical gate; however, there are quantum gates that have no classical counter-part. They act on a set of qubits and, changing their state. That is, if $U$ is a quantum gate and $|q\rangle$ is a qubit, then acting the gate $U$ on the qubit $|q\rangle$ transforms the qubit as follows:

$$|q\rangle \xrightarrow{U} U|q\rangle. \tag{1}$$

# Quantum circuits

Quantum circuits are diagrammatic representations of quantum algorithms. The horizontal dimension corresponds to time; moving left to right corresponds to forward motion in time. They consist of a set of qubits $|q_n\rangle$ which are stacked vertically on the left-hand side of the diagram. Lines, called quantum wires, extend horizontally to the right from each qubit, representing its state moving forward in time. Additionally, they contain a set of quantum gates that are applied to the quantum wires. Gates are applied chronologically, left to right.

# Single-Qubit Gates

A single-qubit gate is a physical action that is applied to one qubit. It can be represented by a matrix $U$ from the group SU(2). Any single-qubit gate can be parameterized by three angles: $\theta$, $\phi$, and $\lambda$ as follows

$$U(\theta, \phi, \lambda) = \begin{bmatrix} \cos\frac{\theta}{2} & -e^{i\lambda}\sin\frac{\theta}{2} \\ e^{i\phi}\sin\frac{\theta}{2} & e^{i(\phi+\lambda)}\cos\frac{\theta}{2} \end{bmatrix}.$$

# Widely used gates

There are several widely used quantum gates. Perhaps the most famous are the Pauli gates correspond to the Pauli matrices

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix},$$

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix},$$

$$Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix},$$

$$Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}.$$

# Algebra basis

These gates form a basis for the algebra $\mathfrak{su}(2)$. Exponentiating them will thus give us a basis for SU(2), the group within which all single-qubit gates live.

# Exponentiated Pauli gates

These exponentiated Pauli gates are called rotation gates $R_\sigma(\theta)$ because they rotate the quantum state around the axis $\sigma = X, Y, Z$ of the Bloch sphere by an angle $\theta$. They are defined as

$$R_X(\theta) = e^{-i\frac{\theta}{2}X} = \begin{bmatrix} \cos\frac{\theta}{2} & -i\sin\frac{\theta}{2} \\ -i\sin\frac{\theta}{2} & \cos\frac{\theta}{2} \end{bmatrix},$$

$$R_Y(\theta) = e^{-i\frac{\theta}{2}Y} = \begin{bmatrix} \cos\frac{\theta}{2} & -\sin\frac{\theta}{2} \\ \sin\frac{\theta}{2} & \cos\frac{\theta}{2} \end{bmatrix},$$

$$R_Z(\theta) = e^{-i\frac{\theta}{2}Z} = \begin{bmatrix} e^{-i\theta/2} & 0 \\ 0 & e^{i\theta/2} \end{bmatrix}.$$

# Basis for $\mathrm{SU}(2)$

Because they form a basis for $\mathrm{SU}(2)$, any single-qubit gate can be decomposed into three rotation gates. Indeed

$$R_z(\phi)R_y(\theta)R_z(\lambda) = \begin{bmatrix} e^{-i\phi/2} & 0 \\ 0 & e^{i\phi/2} \end{bmatrix} \begin{bmatrix} \cos\frac{\theta}{2} & -\sin\frac{\theta}{2} \\ \sin\frac{\theta}{2} & \cos\frac{\theta}{2} \end{bmatrix} \begin{bmatrix} e^{-i\lambda/2} & 0 \\ 0 & e^{i\lambda/2} \end{bmatrix}$$

which we can rewite as

$$e^{-i(\phi+\lambda)/2} \begin{bmatrix} \cos\frac{\theta}{2} & -e^{i\lambda}\sin\frac{\theta}{2} \\ e^{i\phi}\sin\frac{\theta}{2} & e^{i(\phi+\lambda)}\cos\frac{\theta}{2} \end{bmatrix},$$

which is, up to a global phase, equal to the expression for an arbitrary single-qubit gate.

# Two-Qubit Gates

A two-qubit gate is a physical action that is applied to two qubits. It can be represented by a matrix $U$ from the group SU(4). One important type of two-qubit gates are controlled gates, which work as follows: Suppose $U$ is a single-qubit gate. A controlled-$U$ gate ($CU$) acts on two qubits: a control qubit $|x\rangle$ and a target qubit $|y\rangle$. The controlled-$U$ gate applies the identity $I$ or the single-qubit gate $U$ to the target qubit if the control gate is in the zero state $|0\rangle$ or the one state $|1\rangle$, respectively.

# Control qubit

The control qubit is not acted upon. This can be represented as follows if

$$CU|xy\rangle = |xy\rangle \text{ if } |x\rangle = |0\rangle.$$

# In matrix form

It is easier to see in a matrix form. It can be written in matrix form by writing it as a superposition of the two possible cases, each written as a simple tensor product

$$CU = |0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes U = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & u_{00} & u_{01} \\ 0 & 0 & u_{10} & u_{11} \end{bmatrix}.$$

# CNOT gate

One of the most fundamental controlled gates is the CNOT gate. It is defined as the controlled-$X$ gate $CX$. It can be written in matrix form as follows:

$$\mathrm{CNOT} = \mathrm{CX} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}.$$

# CX gate

It changes, when operating on a two-qubit state where the first qubit is the control qubit and the second qubit is the target qubit, the states (check this)

$$\mathrm{CX}|00\rangle = |00\rangle,$$

$$\mathrm{CX}|10\rangle = |11\rangle,$$

$$\mathrm{CX}|01\rangle = |01\rangle,$$

$$\mathrm{CX}|11\rangle = |10\rangle,$$

which you can easily see by simply multiplying the above matrix with any of the above states.

# Swap gate

A widely used two-qubit gate that goes beyond the simple controlled function is the SWAP gate. It swaps the states of the two qubits it acts upon

$$\mathrm{SWAP}|xy\rangle = |yx\rangle.$$

and has the following matrix form

$$\mathrm{SWAP} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

# Simple code

```
#!/usr/bin/env python
# coding: utf-8
import numpy as np
import qiskit as qk
from scipy.optimize import minimize


# # Initialize registers and circuit

n_qubits = 1 #Number of qubits
n_cbits = 1 #Number of classical bits (the number of qubits you want t
qreg = qk.QuantumRegister(n_qubits) #Create a quantum register
creg = qk.ClassicalRegister(n_cbits) #Create a classical register
circuit = qk.QuantumCircuit(qreg,creg) #Create your quantum circuit


circuit.draw() #Draw circuit. It is empty
```

Thereafter we perform operations on qubit

```
circuit.x(qreg[0]) #Applies a Pauli X gate to the first qubit in the q
circuit.draw()
```

and select a qubit to measure and encode the results to a classical bit

```
#Measure the first qubit in the quantum register
#and encode the results to the first qubit in the classical register
```

# Exercises on Bell states

```
import qiskit as qk
from qiskit import QuantumRegister, ClassicalRegister, QuantumCircuit
from qiskit.visualization import plot_histogram
from qiskit_aer import AerSimulator
import matplotlib.pyplot as plt
```

## Setting up a circuit

```
qreg_q = QuantumRegister(2) # 2 qubits
creg_c = ClassicalRegister(2) # 2 classical bits

qc = QuantumCircuit(qreg_q, creg_c) # alternatively, circuit = Quantum

qc.h(qreg_q[0]) # apply the hadamard gate to the first qubit (to the r

# apply the CNOT gate with the first qubit being the control qubit and
qc.cx(qreg_q[0], qreg_q[1])


# measure the qubits, specify which qubit you want to measure
qc.measure(qreg_q[0], creg_c[0])
qc.measure(qreg_q[1], creg_c[1])
```

Next we will use an ideal(noiseless) simulator (AerSimulator) to run our circuit. We will use the qasm_simulator backend. This simulator returns a result object which contains the counts, or

# Testing the Bell states

```
from qiskit import QuantumCircuit, transpile, assemble, IBMQ
from qiskit.visualization import circuit_drawer
import qiskit_aer
import matplotlib.pyplot as plt

# Create a quantum circuit with two qubits
bell_circuit = QuantumCircuit(2, 2)

# Apply Hadamard gate to the first qubit
bell_circuit.h(0)

# Apply a CNOT gate with the first qubit as the control and the second
bell_circuit.cx(0, 1)

# Add measurements to the circuit
bell_circuit.measure([0, 1], [0, 1])

# Visualize the circuit
print("Quantum Circuit:")
print(bell_circuit)
```

# Running the calculations

```
# Number of shots
num_shots = 10000

# Simulate the circuit using the Aer simulator
simulator = qiskit_aer.Aer.get_backend('qasm_simulator')

# Transpile the circuit for the simulator
transpiled_circuit = transpile(bell_circuit, simulator)

# Execute the transpiled circuit on the simulator with the specified n
result = simulator.run(transpiled_circuit, shots=num_shots).result()

# Get measurement counts
counts = result.get_counts(bell_circuit)

# Get and print the measurement results
counts = result.get_counts(bell_circuit)
print("\nMeasurement Results:")
print(counts)
```

# Then plotting the histogram over counts

```
# Plot the histogram using Matplotlib
plt.bar(counts.keys(), counts.values())
plt.title('Bell State Measurement Results')
plt.ylabel('Counts')
plt.show()
```

# Getting serious, Quantum Simulator with Hardware noise model

```
from qiskit import QuantumCircuit, transpile, assemble, Aer, IBMQ
from qiskit.visualization import plot_histogram
from qiskit.providers.aer.noise import NoiseModel

# you will not be able to run unless you provide your token here
api_token = 'your_api_token'
# Load your IBM Quantum account
IBMQ.save_account(api_token, overwrite=True)  # This stores your crede
IBMQ.load_account()
# Get the least busy backend
provider = IBMQ.get_provider(hub='ibm-q', group ='open', project='main
#backend = provider.get_backend('your_preferred_backend')
backend = provider.get_backend('ibm_osaka')

# Get the noise model for the selected backend
noise_model = NoiseModel.from_backend(backend)

# Create a quantum circuit with two qubits
bell_circuit = QuantumCircuit(2, 2)

# Apply Hadamard gate to the first qubit
bell_circuit.h(0)

# Apply a CNOT gate with the first qubit as the control and the second
bell_circuit.cx(0, 1)
```

# Then run and visualize it

```
# Number of shots
num_shots = 10000

# Simulate the circuit using the Aer simulator with the noise model
simulator = Aer.get_backend('qasm_simulator')
noisy_transpiled_circuit = transpile(bell_circuit, simulator, optimiza
noisy_circuit = assemble(noisy_transpiled_circuit, shots=num_shots, no

# Execute the noisy circuit on the simulator
result = simulator.run(noisy_circuit).result()

# Get and print the measurement results
counts = result.get_counts(bell_circuit)
print("\nMeasurement Results:")
print(counts)
# Plot the histogram using Matplotlib
plot_histogram(counts, title='Bell State Measurement Results')
```

# And without the noise model

```
# Provide your IBM Quantum Experience API token here
api_token = 'your_api_token'
# Load your IBM Quantum account
IBMQ.save_account(api_token, overwrite=True)  # This stores your crede
IBMQ.load_account()

# Get the least busy backend
#provider = IBMQ.get_provider(hub='your_hub', group='your_group', proj
provider = IBMQ.get_provider(hub='ibm-q', group ='open', project='main
#backend = provider.get_backend('your_preferred_backend')
backend = provider.get_backend('ibm_osaka')

# Create a quantum circuit with two qubits
bell_circuit = QuantumCircuit(2, 2)

# Apply Hadamard gate to the first qubit
bell_circuit.h(0)

# Apply a CNOT gate with the first qubit as the control and the second
bell_circuit.cx(0, 1)

# Add measurements to the circuit
bell_circuit.measure([0, 1], [0, 1])

# Transpile the circuit for the chosen backend
transpiled_circuit = transpile(bell_circuit, backend)

# Execute the transpiled circuit on the IBM Quantum computer
```

## Gates, the whys and hows

In quantum computing it is common to rewrite various unitary transformations acting on a given state, in terms of so-called gates (one-qubit, two-qubit or more qubit gates). These unitary transformations do actually represent specific interactions of the system with the environment.
Each such operation is by convention written in terms of gates and a chain of such gates represents a circuit. The latter represents then a specific set of operations on an initial state in order to perform what we will label as experiments.
The aim of the first set of notes this week is to link these gates (and thereby circuits) to their respective unitary transformation. These unitary transformations represent selected physical processes.

# Mathematical background

The time-dependent Schrödinger equation (or equation of motion) reads

$$\imath\hbar\frac{\partial}{\partial t}|\Psi_S(t)\rangle = \hat{H}\Psi_S(t)\rangle,$$

where the subscript $S$ stands for Schrödinger here. A formal solution is given by

$$|\Psi_S(t)\rangle = \exp\left(-\imath\hat{H}(t-t_0)/\hbar\right)|\Psi_S(t_0)\rangle.$$

The Hamiltonian $\hat{H}$ is hermitian and the exponent represents a unitary operator with an operation carried ut on the wave function at a time $t_0$.

# Interaction picture

Our Hamiltonian is normally written out as the sum of an unperturbed part $\hat{H}_0$ and an interaction part $\hat{H}_I$, that is

$$\hat{H} = \hat{H}_0 + \hat{H}_I.$$

In general we have $[\hat{H}_0, \hat{H}_I] \neq 0$ since $[\hat{T}, \hat{V}] \neq 0$. We wish now to define a unitary transformation in terms of $\hat{H}_0$ by defining

$$|\Psi_I(t)\rangle = \exp(\imath \hat{H}_0 t/\hbar)|\Psi_S(t)\rangle,$$

which is again a unitary transformation carried out now at the time $t$ on the wave function in the Schrödinger picture.

# Interaction picture

We can easily find the equation of motion by taking the time derivative

$$\imath\hbar\frac{\partial}{\partial t}|\Psi_I(t)\rangle = -\hat{H}_0 \exp\left(\imath\hat{H}_0 t/\hbar\right)\Psi_S(t)\rangle + \exp\left(\imath\hat{H}_0 t/\hbar\right)\imath\hbar\frac{\partial}{\partial t}\Psi_S(t)\rangle.$$

# Interaction picture

Using the definition of the Schrödinger equation, we can rewrite the last equation as

$$\imath\hbar\frac{\partial}{\partial t}|\Psi_I(t)\rangle = \exp\left(\imath\hat{H}_0 t/\hbar\right)\left[-\hat{H}_0 + \hat{H}_0 + \hat{H}_I\right]\exp\left(-\imath\hat{H}_0 t/\hbar\right)\Psi_I(t)\rangle,$$

which gives us

$$\imath\hbar\frac{\partial}{\partial t}|\Psi_I(t)\rangle = \hat{H}_I(t)\Psi_I(t)\rangle,$$

with

$$\hat{H}_I(t) = \exp\left(\imath\hat{H}_0 t/\hbar\right)\hat{H}_I\exp\left(-\imath\hat{H}_0 t/\hbar\right).$$

# Interaction picture

The order of the operators is important since $\hat{H}_0$ and $\hat{H}_I$ do generally not commute. The expectation value of an arbitrary operator in the interaction picture can now be written as

$$\langle \Psi'_S(t)|\hat{O}_S|\Psi_S(t)\rangle = \langle \Psi'_I(t)|\exp\left(\imath \hat{H}_0 t/\hbar\right)\hat{O}_I \exp\left(-\imath \hat{H}_0 t/\hbar\right)|\Psi_I(t)\rangle,$$

and using the definition

$$\hat{O}_I(t) = \exp\left(\imath \hat{H}_0 t/\hbar\right)\hat{O}_I \exp\left(-\imath \hat{H}_0 t/\hbar\right),$$

we obtain

$$\langle \Psi'_S(t)|\hat{O}_S|\Psi_S(t)\rangle = \langle \Psi'_I(t)|\hat{O}_I(t)|\Psi_I(t)\rangle,$$

stating that a unitary transformation does not change expectation values!

# Interaction picture

If the take the time derivative of the operator in the interaction picture we arrive at the following equation of motion

$$\imath\hbar\frac{\partial}{\partial t}\hat{O}_I(t) = \exp\left(\imath\hat{H}_0 t/\hbar\right)\left[\hat{O}_S\hat{H}_0 - \hat{H}_0\hat{O}_S\right]\exp\left(-\imath\hat{H}_0 t/\hbar\right) = \left[\hat{O}_I(t),\hat{H}_0\right]$$

Here we have used the time-independence of the Schrödinger equation together with the observation that any function of an operator commutes with the operator itself.

# Interaction picture

In order to solve the equation of motion equation in the interaction picture, we define a unitary operator time-development operator $\hat{U}(t, t')$. Later we will derive its connection with the linked-diagram theorem, which yields a linked expression for the actual operator. The action of the operator on the wave function is

$$|\Psi_I(t)\rangle = \hat{U}(t, t_0)|\Psi_I(t_0)\rangle,$$

with the obvious value $\hat{U}(t_0, t_0) = 1$.

# Interaction picture

The time-development operator $U$ has the properties that

$$\hat{U}^\dagger(t,t')\hat{U}(t,t') = \hat{U}(t,t')\hat{U}^\dagger(t,t') = 1,$$

which implies that $U$ is unitary

$$\hat{U}^\dagger(t,t') = \hat{U}^{-1}(t,t').$$

Further,

$$\hat{U}(t,t')\hat{U}(t't'') = \hat{U}(t,t'')$$

and

$$\hat{U}(t,t')\hat{U}(t',t) = 1,$$

which leads to

$$\hat{U}(t,t') = \hat{U}^\dagger(t',t).$$

# Interaction picture

Using our definition of Schrödinger's equation in the interaction picture, we can then construct the operator $\hat{U}$. We have defined

$$|\Psi_I(t)\rangle = \exp(\imath \hat{H}_0 t/\hbar)|\Psi_S(t)\rangle,$$

which can be rewritten as

$$|\Psi_I(t)\rangle = \exp(\imath \hat{H}_0 t/\hbar)\exp(-\imath \hat{H}(t - t_0)/\hbar)|\Psi_S(t_0)\rangle,$$

or

$$|\Psi_I(t)\rangle = \exp(\imath \hat{H}_0 t/\hbar)\exp(-\imath \hat{H}(t - t_0)/\hbar)\exp(-\imath \hat{H}_0 t_0/\hbar)|\Psi_I(t_0)\rangle.$$

# Interaction picture

From the last expression we can define

$$\hat{U}(t, t_0) = \exp\left(\imath \hat{H}_0 t/\hbar\right) \exp\left(-\imath \hat{H}(t - t_0)/\hbar\right) \exp\left(-\imath \hat{H}_0 t_0/\hbar\right).$$

It is then easy to convince oneself that the properties defined above are satisfied by the definition of $\hat{U}$.

# Interaction picture

We derive the equation of motion for $\hat{U}$ using the above definition. This results in

$$\imath\hbar\frac{\partial}{\partial t}\hat{U}(t,t_0) = \hat{H}_I(t)\hat{U}(t,t_0),$$

which we integrate from $t_0$ to a time $t$ resulting in

$$\hat{U}(t,t_0) - \hat{U}(t_0,t_0) = \hat{U}(t,t_0) - 1 = -\frac{\imath}{\hbar}\int_{t_0}^{t} dt'\hat{H}_I(t')\hat{U}(t',t_0),$$

which can be rewritten as

$$\hat{U}(t,t_0) = 1 - \frac{\imath}{\hbar}\int_{t_0}^{t} dt'\hat{H}_I(t')\hat{U}(t',t_0).$$

# Interaction picture

We can solve this equation iteratively keeping in mind the time-ordering of the of the operators

$$\hat{U}(t,t_0) = 1 - \frac{\imath}{\hbar}\int_{t_0}^{t} dt'\,\hat{H}_I(t') + \left(\frac{-\imath}{\hbar}\right)^2 \int_{t_0}^{t} dt' \int_{t_0}^{t'} dt''\,\hat{H}_I(t')\hat{H}_I(t'') + \ldots$$

The third term can be written as

$$\int_{t_0}^{t} dt' \int_{t_0}^{t'} dt''\,\hat{H}_I(t')\hat{H}_I(t'') = \frac{1}{2}\int_{t_0}^{t} dt' \int_{t_0}^{t'} dt''\,\hat{H}_I(t')\hat{H}_I(t'') + \frac{1}{2}\int_{t_0}^{t} dt'' \int_{t}$$

# Interaction picture

We obtain this expression by changing the integration order in the second term via a change of the integration variables $t'$ and $t''$ in

$$\frac{1}{2}\int_{t_0}^{t} dt' \int_{t_0}^{t'} dt'' \hat{H}_I(t')\hat{H}_I(t'').$$

We can rewrite the terms which contain the double integral as

$$\int_{t_0}^{t} dt' \int_{t_0}^{t'} dt'' \hat{H}_I(t')\hat{H}_I(t'') =$$

$$\frac{1}{2}\int_{t_0}^{t} dt' \int_{t_0}^{t'} dt'' \left[\hat{H}_I(t')\hat{H}_I(t'')\Theta(t'-t'') + \hat{H}_I(t')\hat{H}_I(t'')\Theta(t''-t')\right],$$

with $\Theta(t''-t')$ being the standard Heavyside or step function. The step function allows us to give a specific time-ordering to the above expression.

# Interaction picture

With the Θ-function we can rewrite the last expression as

$$\int_{t_0}^{t} dt' \int_{t_0}^{t'} dt'' \hat{H}_I(t')\hat{H}_I(t'') = \frac{1}{2}\int_{t_0}^{t} dt' \int_{t_0}^{t'} dt'' \hat{T}\left[\hat{H}_I(t')\hat{H}_I(t'')\right],$$

where $\hat{T}$ is the so-called time-ordering operator.

# Interaction picture

With this definition, we can rewrite the expression for $\hat{U}$ as

$$\hat{U}(t, t_0) = \sum_{n=0}^{\infty} \left(\frac{-\imath}{\hbar}\right)^n \frac{1}{n!} \int_{t_0}^{t} dt_1 \cdots \int_{t_0}^{t} dt_N \, \hat{T}\left[\hat{H}_I(t_1) \ldots \hat{H}_I(t_n)\right] = \hat{T} \exp$$

The above time-evolution operator in the interaction picture will be used to derive various contributions to many-body perturbation theory. See also exercise 26 for a discussion of the various time orderings.

# Interaction picture

We wish now to define a unitary transformation in terms of $\hat{H}$ by defining

$$|\Psi_H(t)\rangle = \exp(\imath\hat{H}t/\hbar)|\Psi_S(t)\rangle,$$

which is again a unitary transformation carried out now at the time $t$ on the wave function in the Schrödinger picture. If we combine this equation with Schrödinger's equation we obtain the following equation of motion

$$\imath\hbar\frac{\partial}{\partial t}|\Psi_H(t)\rangle = 0,$$

meaning that $|\Psi_H(t)\rangle$ is time independent. An operator in this picture is defined as

$$\hat{O}_H(t) = \exp(\imath\hat{H}t/\hbar)\hat{O}_S\exp(-\imath\hat{H}t/\hbar).$$

# Interaction picture

The time dependence is then in the operator itself, and this yields in turn the following equation of motion

$$\imath\hbar\frac{\partial}{\partial t}\hat{O}_H(t) = \exp\left(\imath\hat{H}t/\hbar\right)\left[\hat{O}_H\hat{H} - \hat{H}\hat{O}_H\right]\exp\left(-\imath\hat{H}t/\hbar\right) = \left[\hat{O}_H(t), \hat{H}\right].$$

We note that an operator in the Heisenberg picture can be related to the corresponding operator in the interaction picture as

$$\hat{O}_H(t) = \exp\left(\imath\hat{H}t/\hbar\right)\hat{O}_S\exp\left(-\imath\hat{H}t/\hbar\right) =$$

$$\exp\left(\imath\hat{H}_I t/\hbar\right)\exp\left(-\imath\hat{H}_0 t/\hbar\right)\hat{O}_I\exp\left(\imath\hat{H}_0 t/\hbar\right)\exp\left(-\imath\hat{H}_I t/\hbar\right).$$

# Interaction picture

With our definition of the time evolution operator we see that

$$\hat{O}_H(t) = \hat{U}(0,t)\hat{O}_I\hat{U}(t,0),$$

which in turn implies that $\hat{O}_S = \hat{O}_I(0) = \hat{O}_H(0)$, all operators are equal at $t = 0$. The wave function in the Heisenberg formalism is related to the other pictures as

$$|\Psi_H\rangle = |\Psi_S(0)\rangle = |\Psi_I(0)\rangle,$$

since the wave function in the Heisenberg picture is time independent. We can relate this wave function to that a given time $t$ via the time evolution operator as

$$|\Psi_H\rangle = \hat{U}(0,t)|\Psi_I(t)\rangle.$$

# Interaction picture

We assume that the interaction term is switched on gradually. Our wave function at time $t = -\infty$ and $t = \infty$ is supposed to represent a non-interacting system given by the solution to the unperturbed part of our Hamiltonian $\hat{H}_0$. We assume the ground state is given by $|\Phi_0\rangle$, which could be a Slater determinant. We define our Hamiltonian as

$$\hat{H} = \hat{H}_0 + \exp(-\varepsilon t/\hbar)\hat{H}_I,$$

where $\varepsilon$ is a small number. The way we write the Hamiltonian and its interaction term is meant to simulate the switching of the interaction.

# Interaction picture

The time evolution of the wave function in the interaction picture is then

$$|\Psi_I(t)\rangle = \hat{U}_\varepsilon(t, t_0)|\Psi_I(t_0)\rangle,$$

with

$$\hat{U}_\varepsilon(t, t_0) = \sum_{n=0}^{\infty} \left(\frac{-\imath}{\hbar}\right)^n \frac{1}{n!} \int_{t_0}^{t} dt_1 \cdots \int_{t_0}^{t} dt_N \exp\left(-\varepsilon(t_1 + \cdots + t_n)/\hbar\right) \hat{T} \left[\right.$$

# Interaction picture

In the limit $t_0 \to -\infty$, the solution ot Schrödinger's equation is $|\Phi_0\rangle$, and the eigenenergies are given by

$$\hat{H}_0|\Phi_0\rangle = W_0|\Phi_0\rangle,$$

meaning that

$$|\Psi_S(t_0)\rangle = \exp\left(-\imath W_0 t_0/\hbar\right)|\Phi_0\rangle,$$

with the corresponding interaction picture wave function given by

$$|\Psi_I(t_0)\rangle = \exp\left(\imath \hat{H}_0 t_0/\hbar\right)|\Psi_S(t_0)\rangle = |\Phi_0\rangle.$$

# Interaction picture

The solution becomes time independent in the limit $t_0 \to -\infty$. The same conclusion can be reached by looking at

$$\imath\hbar\frac{\partial}{\partial t}|\Psi_I(t)\rangle = \exp\left(\varepsilon|t|/\hbar\right)\hat{H}_I|\Psi_I(t)\rangle$$

and taking the limit $t \to -\infty$. We can rewrite the equation for the wave function at a time $t = 0$ as

$$|\Psi_I(0)\rangle = \hat{U}_\varepsilon(0, -\infty)|\Phi_0\rangle.$$